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RESEARCH ARTICLE

Enhancing Quantum Cryptanalysis of Binary Elliptic Curves Through Optimized Out-of-Place Point Addition and ECPM Integration

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ABSTRACT Shor's algorithm remains a cornerstone of quantum cryptanalysis, driving active research into reducing quantum resource overheads in both spatial and temporal dimensions. This work presents a dual-optimization framework targeting resource-efficient Shor-based quantum cryptanalysis on binary elliptic curves. Our approach conducts a comprehensive resource assessment across multiple abstraction levels—ranging from finite field arithmetic to elliptic curve group operations. At the low-level layer, we benchmark various quantum arithmetic constructor designs—including GCD-based, FLT-based, FLT-Basic or extended variants, in-place, and out-of-place execution models—analyzing Toffoli gate counts, qubit requirements, and circuit depths. At the group operation layer, we examine point addition and doubling circuits, employing complete iterations of Shor's algorithm with elliptic curve point multiplication (ECPM) that require point addition (PA) only (with $(2n + 2)$ PA) and point doubling (PD) utilization ($2n$ PD + 2 PA). We further simulate selected $(2n$ PD + 2 PA_{outplace}) circuit designs, highlighting the advantages of optimized out-of-place PA circuits and integrated ECPM techniques from prior work. Additionally, we explore decompositions of multi-controlled gates to achieve a practical trade-off between qubit count and overall depth.

INDEX TERMS Elliptic curve, ECPM, quantum cryptanalysis, Shor algorithm.

I. INTRODUCTION

In the domain of information security, cryptanalysis refers to the systematic examination of cryptographic protocols and algorithms with the goal of uncovering structural weaknesses or potential vulnerabilities. Classically, this has involved extensive computational efforts using brute-force or

statistical methods. However, the advent of quantum computing has fundamentally altered this landscape, enabling more efficient attacks through quantum-enhanced algorithmic strategies [1]. Notably, Shor's algorithm [2] introduces a polynomial-time solution to problems such as integer factorization and discrete logarithms, both of which underlie many conventional cryptographic systems. In parallel, Grover's algorithm [3] provides quadratic speedups in unstructured search, further extending the reach of quantum cryptanalysis.

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Among the cryptographic schemes potentially threatened by quantum capabilities, Elliptic Curve Cryptography (ECC) stands out due to its widespread adoption in secure communications. Introduced independently by Koblitz and Miller in the mid-1980s, ECC offers high security with significantly smaller key sizes compared to RSA and other legacy systems. This key-size efficiency has led to its integration into modern protocols such as TLS 1.3 [8], and strong endorsements in standards and recommendations from institutions like NIST [9]. The cryptographic strength of ECC relies on the difficulty of solving the Elliptic Curve Discrete Logarithm Problem (ECDLP).

There is a growing need for methodologies that address both spatial (width) and temporal (depth) complexities simultaneously. An exemplary instance of ongoing research in resource optimization (e.g., qubit count, overall depth) is Shor-based quantum cryptanalysis designed for elliptic curve cryptographic systems. This dual-efficiency perspective is particularly critical for Elliptic Curve over binary fields, where finite field arithmetic operations—such as inversion, multiplication, and squaring—must be tightly optimized to manage the trade-offs between accuracy, width, and depth.

A number of quantum studies—including those by Roetteler et al. [10] and Häner et al. [11]—have extended Shor's original framework to target ECDLP specifically. These efforts have led to increasingly efficient quantum circuit constructions tailored for elliptic curve cryptanalysis. More recently, as shown in [5], [7], [12], [13], and [14], research has intensified around optimizing these circuits, particularly focusing on reducing both the qubit count and circuit depth.

In this work, we present a comprehensive space- and time-efficient quantum cryptanalysis framework for binary elliptic curves, grounded in a full implementation pipeline of elliptic curve hierarchy. Our analysis covers multiple hierarchical levels—from low-level field arithmetic to high-level point operations—benchmarked across various schemes, such as Greatest Common Divisor (GCD)-based and Fermat's Little Theorem (FLT)-based modular arithmetic and includes both in-place and out-of-place implementations. The resulting resource profiles provide key insights into each level's progress to the overall quantum cost of elliptic curve cryptanalysis, ultimately guiding future optimization strategies and secure cryptographic design in the post-quantum era. The key contributions are as follows:

- 1) We analyze hierarchical implementations of quantum cryptanalysis targeting binary elliptic curve circuit designs, with a focus on both existing methods and optimization layers in field arithmetic over $\text{GF}(2^m)$. Additionally, we propose improvement strategies at each hierarchy level, contextualizing recent advances in quantum circuit construction within the framework of Shor's algorithm for solving the ECDLP over binary fields.
- 2) We perform an in-depth comparative analysis of Level-1 circuit designs, evaluating various inversion and multiplication techniques in terms of Toffoli gate count,

qubit requirements, and circuit depth. This includes benchmarking FLT-based inversion with Karatsuba-depth optimization [5], GCD-based and FLT-based designs [12], and basic vs. extended FLT-based inversion approaches [13], as well as in-place and out-of-place arithmetic strategies [14]. To validate scalability and practical integration, we simulate selected configurations using the Qiskit and ETRI QCrypton frameworks and provide a preliminary analysis of quantum error overheads using a Quantum Software Development Kit (SDK) environment.

- 3) We examine binary elliptic curve Group operation (e.g., Point Addition (PA), Point Doubling (PD)) as well as key scalar multiplication components—namely elliptic curve point multiplication (ECPM)—with a setup that utilizes point additions only (with $2n + 2$ PA) and point doubling utilization ($2n$ PD + 2 PA). Designs from [5], [7], [12], [14], and [15] are assessed. We simulate these configurations using Qiskit to demonstrate their scalability and integration potential within Shor-based quantum cryptanalysis.
- 4) We present a resource-efficiency analysis that highlights trade-offs between circuit depth (time) and qubit count (space) across representative designs. Our benchmarking identifies configurations that strike an effective balance between these factors, focusing on optimized out-of-place point addition, efficient ECPM integration, and multigate cutting techniques.

This paper is organized as follows. Section I introduces the background and outlines the primary contributions of the study. Section II reviews relevant literature and presents the proposed hierarchical structure for quantum ECC implementations, categorized from Level 1 to Level 4. Section III examines recent advancements and state-of-the-art quantum circuit designs, with detailed analysis of quantum cost and curve-specific visualizations at each hierarchical level. Section IV presents simulation results and discusses optimization strategies aimed at achieving space- and time-efficient quantum cryptanalysis. Finally, Section V concludes the paper by summarizing the key findings and future research directions.

II. RELATED WORK

This work focuses on a hierarchical elaboration of elliptic curves defined over binary fields $\mathbb{F}_{2^n}$, building upon methodologies discussed in prior studies ([5], [6], [7], [12], [13], [14]) to ensure a valid and consistent basis for comparison. We illustrate the wide range of approaches at each implementation level in Figure 1, and highlight the specific differences in taxonomy and implementation details in Table 2.

Each field element in $\mathbb{F}_{2^n}$ is represented as a polynomial of degree less than n with coefficients in $\mathbb{F}_2$. Arithmetic operations are performed via the isomorphic representation $\mathbb{F}_{2^n} \cong \mathbb{F}_2[z]/m(z)$, where $m(z)$ denotes an irreducible polynomial of degree n over $\mathbb{F}_2[z]$. All computations are

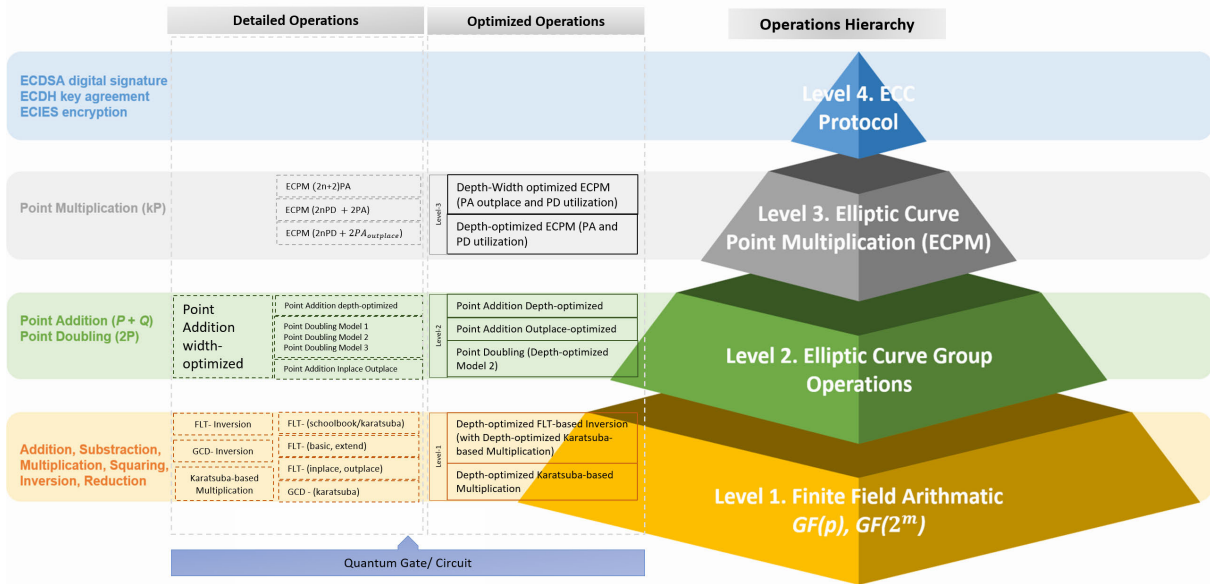


FIGURE 1. General Hierarchy in ECC Implementation. The base structure of the hierarchy in this figure is redrawn from [4], [5], [6], and [7]. The detailed operation annotations indicate the type of elaboration relevant to each level. The figure highlights existing depth-width optimization strategies at each stage, along with the present proposed approach, with particular emphasis on Level 3 ($2n$ PD + 2 PA_{outplace}) circuit design for constructing an efficient Shor-based elliptic curve quantum cryptanalysis framework.

conducted modulo $m(z)$, enabling structured and scalable implementations of field operations within the context of Shor-based quantum cryptanalysis. In addition to this standard polynomial representation of $\mathbb{F}_{2^n}$, we also incorporate a curated table of irreducible polynomials of degree n (see Table 1), ensuring a fair and consistent baseline for comparison with previous approaches.

TABLE 1. Standardized Irreducible Polynomials. A selection of irreducible polynomials commonly used to define binary extension fields $\mathbb{F}_{2^n}$ in elliptic curve cryptography (ECC) implementations is presented. These polynomials have been adopted in several prior studies, including those by Banegas et al. [12], Putranto et al. [5], and Wardhani et al. [7], and are also utilized throughout this work.

Degree (n)	Irreducible polynomial	Source
8	$z^8 + z^4 + z^3 + z + 1$	[16]
16	$z^{16} + z^5 + z^3 + z + 1$	[16]
127	$z^{127} + z + 1$	[16]
163	$z^{163} + z^7 + z^6 + z^3 + 1$	[17]
233	$z^{233} + z^{74} + 1$	[17]
283	$z^{283} + z^{12} + z^7 + z^5 + 1$	[17]
571	$z^{571} + z^{10} + z^5 + z^2 + 1$	[17]

A binary elliptic curve over $\mathbb{F}_{2^n}$ is described by the equation $y^2 + xy = x^3 + ax^2 + b$, where a is an element of $\mathbb{F}_2$ and element non-zero b belongs to the multiplicative group $\mathbb{F}_{2^n}^*$. Curve points take the form $P = (x, y) \in \mathbb{F}_{2^n}^2$ and must satisfy the curve equation. The identity element, denoted O , represents the point at infinity and functions as the neutral element for the elliptic curve addition operation. The inverse of a point $P_1 = (x_1, y_1)$ is given by $-P_1 = (x_1, y_1 + x_1)$, ensuring that $P_1 + (-P_1) = O$ holds.

Point addition (PA) between two distinct points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2) \neq \pm P_1$ yields a point

$P_1 + P_2 = P_3 = (x_3, y_3)$, computed as:

$$\lambda = \frac{y_1 + y_2}{x_1 + x_2}, \quad x_3 = \lambda^2 + \lambda + x_1 + x_2 + a, \\ y_3 = (x_2 + x_3)\lambda + x_3 + y_2.$$

Point doubling (PD), where P_1 is not equal to its inverse $-P_1$, results in $[2]P_1 = (x_3, y_3)$ using the following formulas:

$$\lambda = x_1 + \frac{y_1}{x_1}, \quad x_3 = \lambda^2 + \lambda + a, \quad y_3 = x_1^2 + (\lambda + 1)x_3.$$

Fundamental elliptic curve operations have been implemented through various algorithmic strategies, with numerous prior studies aiming to optimize them for lower quantum resource usage—particularly when integrated into classical simulations (e.g. hardware-based processors [18], [19], [20]), or quantum computing platforms [21], [22], [23], [24], as illustrated in the comparative analysis in Table 2. These studies provide resource estimations and emphasize the practical adaptation of elliptic curve hierarchical implementation in a quantum environment.

In this work, as part of a comparative implementation of existing arithmetic-level (Level 1) components, PA variants, and the inclusion of PD within a Shor-based quantum cryptanalysis framework for binary elliptic curves, we review and summarize various improvements by organizing them into a hierarchical structure, as illustrated in Figure 1. The figure presents a layered design commonly employed in elliptic curve implementations.

At the base (Level 1) are fundamental finite field arithmetic operations over either $\text{GF}(p)$ or $\text{GF}(2^m)$; in this study, we focus specifically on binary field arithmetic over $\text{GF}(2^m)$,

with particular emphasis on the Division (D), Multiplication (M) and Squaring (S) from components used in the arithmetic circuits of modules (e.g. [5], [12], and [13]). These serve as the foundational building blocks for higher-level operations.

Level 2 builds upon this foundation by forming circuits for point addition (PA, [5], [12], [13] and [14]) and point doubling (PD [15]), which are essential operations in elliptic curve arithmetic. Finally, Level 3 integrates these components to implement elliptic curve point multiplication (ECPM, [6], [7]). This layered analysis provides a structured perspective on the range of optimization strategies employed in prior works—not only those focusing on reducing circuit width or depth individually, but also those aiming to optimize both simultaneously. Our objective is to support the development of more space- and time-efficient quantum cryptanalysis techniques.

Building on Shor [2] for solving discrete logarithms in finite fields, Proos and Zalka adapted Shor's discrete logarithm algorithm for elliptic curves, offering a version tailored to quantum circuit frameworks for certain elliptic curve groups [25]. This adaptation spurred further developments aimed at the practical realization of quantum cryptanalysis on ECC. Notable contributions include circuit-level designs by Roetteler et al. [10], followed by improvements from Haner et al. [11], both targeting ECC over prime fields.

Parallel to these developments, research into quantum cryptanalysis binary elliptic curves has increased. Banegas et al. [12] contributed significantly in this direction by focusing on concrete realizations of Shor's algorithm applied to binary elliptic curves. They employed Van Hoof's variant of Karatsuba multiplication [26], which strikes a balance between gate count and circuit width, offering a more space-efficient construction.

Complementing earlier studies on quantum cryptanalysis over binary fields for elliptic curves, such as the approach by Banegas et al. [12], subsequent works by Putranto et al. [5] and Larasati et al. [4] have further advanced depth-optimization techniques. To provide a clearer context for the proposed enhancement—Quantum Cryptanalysis of Binary Elliptic Curves through Optimized Out-of-Place Point Addition and ECPM Integration—this paper presents a structured analysis to identify existing designs and highlight the proposed improvements. The discussion is organized according to a hierarchical framework for elliptic curve implementations, as illustrated in Figure 1. Numerous follow-up studies have since expanded upon both depth and width optimization across various layers of ECC implementation, as summarized in mapping techniques for level-based elaboration in hierarchical elliptic curve implementations (Table 2).

Further on the Shor-based quantum circuit for binary elliptic curve development, several investigations targeted Level 1 (arithmetic layer) with optimizations such as FLT-based inversion [4] and improved Karatsuba

multiplication [5], [26], as well as Level 2 with three different models of PD [15], including depth-optimized PA designs [5]. These studies were further extended by Putranto et al. [6] and Wardhani et al. [7], providing a comparative analysis at Level 3 (ECPM).

Furthermore, this line of research was deepened by the notable work of Taguchi et al. [13], which applied an addition chain approach, and was later compiled and expanded upon in the quantum cryptanalysis framework developed by Jang et al. [14]. This growing body of work reflects the ongoing pursuit of practical and efficient quantum attacks against ECC by minimizing quantum cost in terms of both width and depth.

III. QUANTUM DESIGN CONSTRUCTORS

The basic unit of quantum computation is the qubit. Unlike classical bits, which can exist only in a definite state of either 0 or 1, qubits can exist in a superposition of the $|0\rangle$ and $|1\rangle$ states. This property enables quantum systems to process a vast number of possibilities in parallel, offering a significant computational advantage.

A qubit's state is typically represented as $\alpha|0\rangle + \beta|1\rangle$, where α and β are complex coefficients satisfying $|\alpha|^2 + |\beta|^2 = 1$. The probability of measuring the qubit in state $|0\rangle$ is $|\alpha|^2$, while that of measuring $|1\rangle$ is $|\beta|^2$. A balanced superposition occurs when $|\alpha| = |\beta|$, for instance, in the state $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$, leading to equal measurement probabilities. When multiple qubits are entangled, the resulting system spans a quantum state space of size 2^n , where n is the number of qubits. This full-state superposition can be written as $\sum_{i=0}^{2^n-1} \alpha_i |q_{n-1,i} q_{n-2,i} \dots q_{1,i} q_{0,i}\rangle$, with $\sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1$, where the binary representation of index i corresponds to the individual qubit states.

In practice, quantum operations are executed on physical qubits implemented through hardware platforms such as superconducting circuits, trapped ions, or photonic systems. However, this study focuses on logical qubits, emphasizing quantum cost estimation through numerical methods—such as interpolation and extrapolation—as well as quantum SDK circuit simulations. In this context, we utilize the Qiskit and QCrypton environments in conjunction with numerical quantum cost analysis techniques. For simulation purposes, Qiskit was employed on a desktop PC with x64 architecture, equipped with an Intel i7-8700 processor (six cores), 64 GB RAM, and running Windows 10 Pro. The software environment consisted of Python 3.9.7 and Qiskit version 0.30.0.

Quantum gates, including CNOT and Toffoli gates, form the building blocks of quantum circuits. These reversible gates allow for the implementation of essential arithmetic operations such as addition, subtraction, multiplication, division, and inversion. These operations are crucial for executing quantum algorithms, particularly those involving elliptic curve arithmetic. These include operations critical to

TABLE 2. Mapping Techniques for Level-Based Elaboration in Hierarchical Elliptic Curve Implementations. This summary integrates insights from both classical and quantum computing literature, organized chronologically by year of publication. It captures recent advancements to enhance the understanding of level-based elliptic curve implementations. It highlights notable contributions, the types of finite fields employed, and the computational frameworks used in prior research. The levels presented in the table correspond to an implementation-level evaluation of elliptic curve operations, aligned with the hierarchical structure depicted in Figure 1.

No	Reference	Elliptic Curve Level-Implementation	Finite Field	Environment/ Platform
1	[27]	Example on Shor's ECDLP (Algorithm-level analysis)	GF ((2^m)) (Example purpose)	Classical
2	[25]	Levels 1-3 (Algorithm-level analysis)	GF (p)	Quantum
3	[21]	Levels 1 and 2	GF (2^m)	Quantum
4	[28]	Level 1	GF (2^m)	Quantum
5	[22]	Level 2	GF (2^m)	Quantum
6	[23]	Level 1	GF (2^m)	Quantum
7	[29]	Level 2	GF (2^m)	Quantum
8	[30]	Levels 1 and 3 (Gate-level)	GF (2^m)	Quantum
10	[18]	Levels 1-3	GF (p)	Classical / Hardware
11	[19]	Levels 1-3	GF (p)	Classical Hardware (FPGA)
12	[24]	Levels 1-2	GF (2^m)	Quantum / Python
13	[10]	Levels 1-3 (Gate-level)	GF (p)	Quantum / LiQuil (F#)
14	[20]	Levels 1-3	GF (p)	Classical Hardware (FPGA)
15	[11]	Levels 1-3 (Gate-level)	GF (p)	Quantum / QDK(Q#)
16	[12]	Levels 1-3 (with PA-only-based PM)	GF (2^m)	Quantum / QDK(Q#)
17	[5]	Levels 1-3 (with PA-only-based PM)	GF (2^m)	Quantum/Qiskit
18	[31]	Levels 1-3	GF (p)	Quantum
19	[32]	Level 1	GF (2^m)	Quantum
20	[7]	Levels 1-3 (with PA and PD-based PM)	GF (2^m)	Quantum/Qiskit
21	[13]	Levels 2	GF (2^m)	Quantum/Qiskit
22	[14]	Levels 1-3 (with improved in-place and out-of-place FLT-based inversion and PA)	GF (2^m)	Quantum/Qiskit
23	Our	Levels 1-3 (with PA (out-of-place design improvement from [14]) and PA/PD-based PM)	GF (2^m)	Quantum/Qiskit

elliptic curve cryptanalysis, such as PA, PD, and multiplication (ECPM).

This study focuses on the design of quantum circuits for binary elliptic curve cryptanalysis, with frequent references to high-level circuit blocks such as inversion, PA, PD, and scalar multiplication. These blocks are composed of various quantum gates and are constructed based on established naming conventions from prior research (e.g., [4], [5], [12], [26]). For instance, the CONSTMODMULT^{-1} block introduced by Putranto et al. [5] performs modular multiplication with constants, while the MODSHIFT^{-1} block is used for modular shift operations. To enhance clarity and assist readers in interpreting the quantum circuits or block schematics used in this extended work, references to specific high-level circuit blocks will be provided throughout the paper.

A. LEVEL-1 BINARY FIELD ARITHMETIC

1) IMPROVED KARATSUBA MULTIPLICATION

Karatsuba multiplication in the quantum environment was implemented by Banegas et al. in [12], incorporating the space-efficient Karatsuba variants introduced by van Hoof [26]. These implementations require $O(n^2)$ CNOT gates, $O(n^{\log_2 3})$ Toffoli gates, and utilize a total of $3n$ qubits. Banegas et al. focused on analyzing and optimizing quantum circuits tailored for computing discrete logarithms over binary elliptic curves, including reversible designs for fixed-base scalar multiplication. Their objective was to minimize the number of logical qubits needed for the practical realization of Shor's algorithm. For an elliptic curve over a field of 2^n elements, their reduced the values to $7n + \lceil \log_2(n) \rceil + 9$ qubits, $O(n^3)$ CNOT gates, and $48n^3 + 8n^{\log_2(3)+1} + 352n^2 \log_2(n) + 512n^2 + O(n^{\log_2(3)})$ Toffoli

gates with double-and-add scalar multiplication [12]. Their work provided concrete estimations of quantum resources needed to solve the discrete logarithm problem using Shor's algorithm on binary elliptic curves, primarily through the optimization of both the multiplication base and division subcircuits.

Subsequent work by Putranto et al. [5], [34] introduced modifications to the computational steps of van Hoof's optimized Karatsuba multiplier. These adjustments aimed to achieve the same Toffoli Karatsuba multiplication outcome while significantly reducing quantum resource consumption—specifically in terms of circuit depth and CNOT gate count. Putranto's design primarily employs the use of binary shift operations (MODSHIFT^{-1}) and efficient relabeling (or free swap) techniques that depend on the structure of the irreducible polynomial rather than the bit-length parameter n degree polynomial.

By altering several key steps from the original algorithm by van Hoof [26], resource costs in Putranto et al. [5], [34] were minimized. The modified approach includes three calls to the KMULT_n subroutine: two of these perform multiplications on $k \times k$ operands, and the third handles a multiplication on $(n-k) \times (n-k)$ operands. Additionally, the method employs $2k$ instances of $\text{MODSHIFT}_m(x)$ to achieve multiplication by x^k , one of which is executed in the reverse direction. A single $\text{CONSTMODMULT}_{f(x),m(x)}$ operation is also included, consistently multiplying by the constant polynomial $1 + x^k$. Finally, the circuit design incorporates four CNOT operations applied to $(n-k)$ qubits, with half of these gates being parallelizable, thereby reducing overall execution depth. Furthermore, Putranto's work restructured the sequence of MODMULT operations from the original implementations in [12] and [26]. Instead of relying on

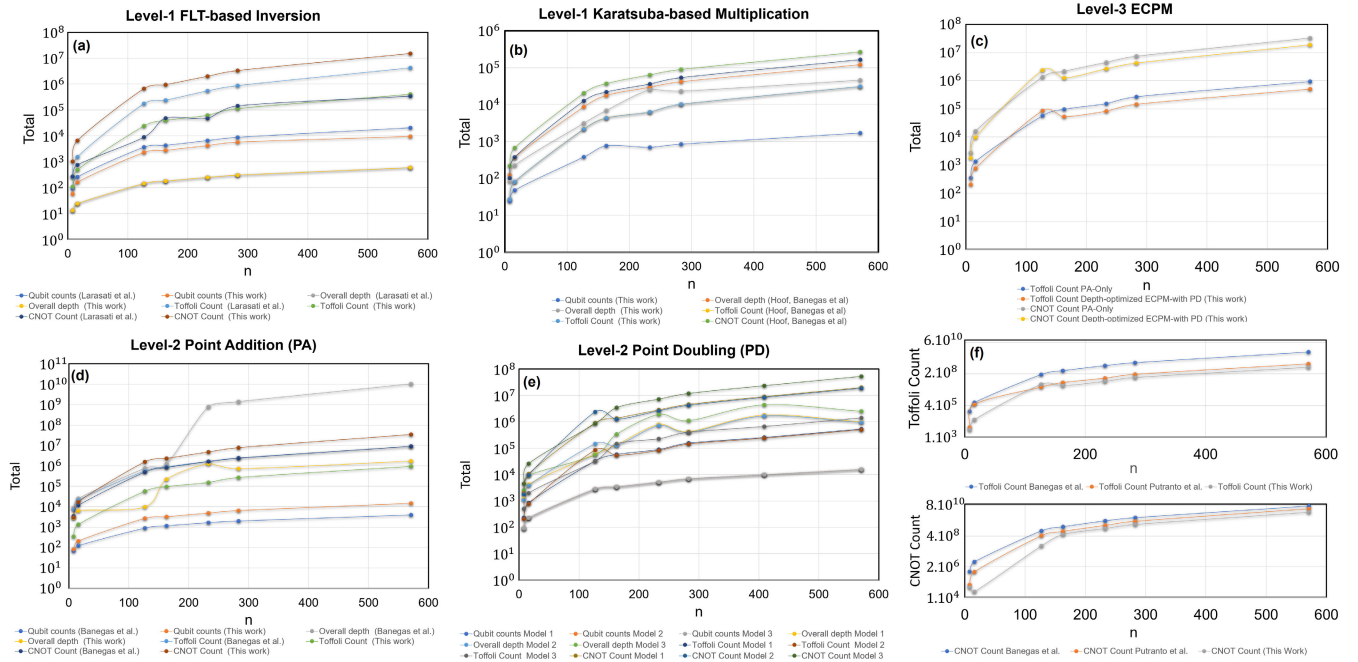


FIGURE 2. Various Design Quantum Cost Comparison at Each Level in Shor-based Binary Elliptic Curve Quantum Cryptanalysis. This figure presents a comprehensive comparative analysis of Level-1 circuit designs, including FLT-based inversion [4] with Karatsuba-depth optimization [5], as shown in Figure (a). Multiplication techniques in Figure (b) include benchmarking of the van Hoof [26]–improved Karatsuba method in Banegas et al. [12], along with depth-optimized designs by Putranto et al. [5]. Figures (c), (d), and (e) present quantum resource comparisons for Level 2–3 designs proposed by Banegas et al. [12], Larasati et al. [15], Putranto et al. [6], and Wardhani et al. [7]. Figure (f) provides a more detailed comparison of Toffoli and CNOT gate counts for single-step Level-3 operations, showcasing the improved ECPM design utilized in this work.

repeated CONSTMODMULT^{-1} calls—which introduce higher depth and CNOT overhead as n grows—the revised design, labeled $\text{MODMULT}_{\text{Imp}}$, offers a more depth-efficient and scalable alternative.

These results serve as the foundation for constructing the Level 3 ECPM circuit in this study. We further elaborate on the quantum resource utilization based on Qiskit simulation results for multiplication, analyzing key metrics including qubit count, overall circuit depth, Toffoli count, and CNOT count. The depth optimization benefits resulting from this approach are demonstrated in Figure 2, specifically in Level-1 Karatsuba implementations shown in subfigure (b).

2) IMPROVED FLT-BASED INVERSION

In the context of Level-1 division in binary field arithmetic, recent advancements have concentrated on enhancing inversion design to reduce quantum cost, particularly with respect to Toffoli gate counts, qubit utilization, and circuit depth. Banegas et al. [12] had initially proposed two quantum circuit designs for implementing Shor’s algorithm in the context of binary elliptic curves, leveraging both GCD- and FLT-based inversion methods. Since then, the FLT-based design has gained wider adoption and refinement across multiple studies.

Larasati et al. [4] introduced a depth-optimized approach to FLT-based inversion over binary fields. Their method employs a “complete waterfall” technique to transform a variant of the Itoh–Tsuji algorithm into a more

quantum-friendly form, effectively removing the need for inverse squaring operations that were present in earlier works such as [12]. The procedure, formalized in Algorithm 1, has become a foundational building block for subsequent designs and utilized in this study. Their optimization significantly reduces the number of CNOT gates and offers a measurable reduction in depth, thus improving the overall quantum resource efficiency. Further refinement incorporates an optimized Karatsuba multiplication technique developed by Putranto et al. [5], and extends the earlier contributions of [12] and [26], yielding a more compact and depth-efficient quantum circuit suitable for scalable implementations.

Notably, Taguchi et al. [13] introduced two inversion strategies—FLT-basic and FLT-extended—which were later further optimized by Jang et al. [14] through both in-place and out-of-place circuit implementations. These efforts collectively represent a growing body of work dedicated to efficient inversion techniques in quantum circuit design. These optimization strategies yield a significant reduction in quantum resource usage—particularly in qubit count and circuit depth—thus improving the practicality of quantum cryptanalysis on binary elliptic curves, as shown in Figure 2(a) quantum cost results comparison.

In addition, this study extends the investigation by evaluating the efficiency of these techniques within an alternative simulation environment: ETRI QCrypton, a quantum circuit framework developed in South Korea. This work utilizes an enhanced version of the platform compared to

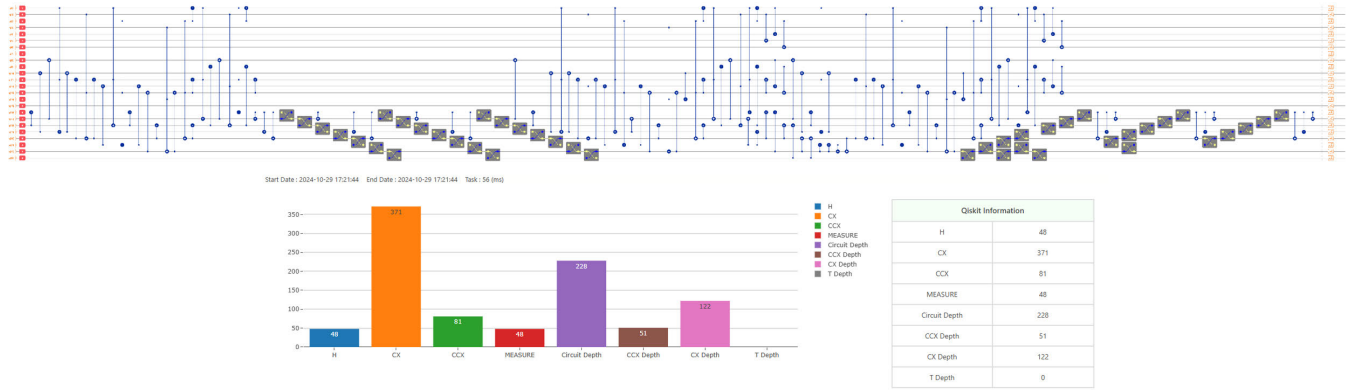


FIGURE 3. Improved Karatsuba Multiplication Quantum circuit Simulation with ETRI QCrypton. This figure shows the simulation result of improved Karatsuba multiplication for a polynomial of degree $n = 8$, representing an example of a Level-1 implementation of arithmetic multiplication using the ETRI QCrypton framework.

Algorithm 1 INV_{Imp} . The FLT-based Inversion Approach Is Based on the Algorithm Introduced by Larasati Et Al. [4], While the Module Labels and Notation Conventions Follow Adaptations From Prior Works Such as [5], [12], [26], and [33].

Fixed input: A constant field polynomial $m(x)$ of degree $n > 0, k_1 > k_2 > \dots k_t \geq 0$

such that $\sum_{s=1}^3 2^{k_s} = n - 1$. $k_{max} = 2k_1 + t$.

Quantum input:

- A non-zero binary polynomials of degree up to $n - 1$ stored in array (register) f_0 of size n to invert.
- k zero arrays of size n initialized to an all- $|0\rangle$ state: $f_1, \dots, f_k$.

Result: inverse of the input, stored in f_k

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1: for  $i = 1, \dots, k_1$  do //stage 1
2:   CNOT( $f_{2(i-1)+1}, f_{2(i-1)}$ )
3:   for  $i = 1, \dots, t - 1$  do
4:     SQUARE( $f_{2(i-1)+1}$ )
5:     MODMULT_Imp( $f_{2(i-1)+2}, f_{2(i-1)+1}, f_{2(i-1)}$ )
6:   for  $s = 1, \dots, t - 1$  do //stage2
7:     for  $k = 1, \dots, 2^{k_s+1}$  do
8:       SQUARE( $f_{2(i-1)+1}$ )
9:       MODMULT_Imp( $f_{2(i-1)+2}, f_{2(i-1)+1}, f_{2(i-1)}$ )
10:    if  $t = 1$  then
11:      swap( $f_{k_1}, f_k$ )
12:    SQUARE( $f_k$ ) //stage 3

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the one employed in [35]. Using QCrypton visualization, we further examine the optimized Level 1 Karatsuba multiplier (Figure 3) and the FLT-based inversion circuit (Figure 4). The results provide a detailed comparative analysis of single-operation quantum gate usage, offering new insights into circuit-level efficiency across different simulation platforms.

B. LEVEL-2 ELLIPTIC CURVE GROUP OPERATIONS

1) ENHANCED OUT-OF-PLACE POINT ADDITION

In the Level 2 implementation, in Figure 5(a) the Point Addition (PA) circuit originally introduced by Banegas et al. [12] has undergone significant refinements, particularly in terms of reducing circuit depth, as further explored by [5], [14], and [34]. A comparison of the quantum cost across each implementation level within Shor-based quantum cryptanalysis over binary elliptic curves—specifically for Point Addition depicted in Figure 2(d).

Building on the analyses provided by Banegas et al. and Putranto et al., the current Jang et al. [14] work further incorporates the method developed, depicted in the Improved Out-of-Place PA ($PA_{outplace}$) scheme Figure 5. Jang et al. [14] focused on optimizing point addition within binary elliptic curve cryptography, with an emphasis on depth minimization. As depicted quantum circuit in Figure 5(c), their most notable contribution is the development of an out-of-place PA strategy that uses additional ancilla qubits to significantly reduce the quantum circuit depth.

This optimized design achieves more than a 92% reduction in the qubit-depth product for a single PA operation compared to previous implementations (e.g., [12], [13], [34]), and shows 73%–81% improvement compared to in-place PA methods. These gains in both space and time efficiency are attributed to optimizations across multiple logical levels: at the component level (e.g., depth-efficient out-of-place squaring circuits and Jang et al.'s multiplication method), at the combination level (through Fermat's Little Theorem (FLT)-based inversion that reuses qubits and leverages reverse operations to maintain shallow circuit depth), and at the architectural level. At this level, two implementations—FLT-in and FLT-out—are presented.

Further, the outcomes of these point addition optimizations—developed respectively by Banegas et al. (BAN) [12] in green, Putranto et al. (PUT) [5] in purple, Wardhani et al. (WAR) [7] in red, and Jang et al. (JAN) [14] in yellow—are consolidated in the final comparative table (Table 3) and graphically represented in Figure 7,

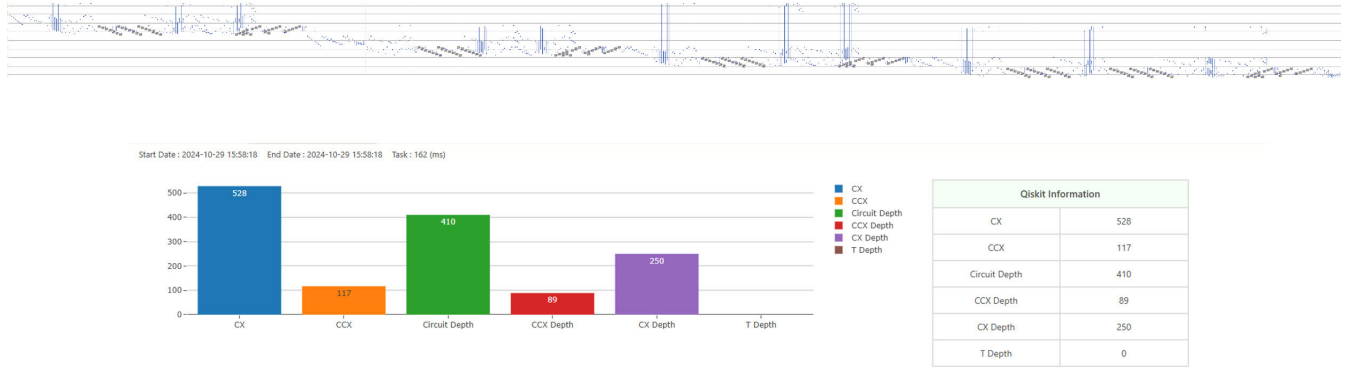


FIGURE 4. FLT-based Inversion Quantum circuit from ETRI QCrypton. This figure shows the simulation result of Inversion Algorithm 1 for a polynomial of degree $n = 8$, representing an example of a Level-1 implementation of FLT-based inversion using the ETRI QCrypton framework.

highlighting the progressive improvements in performance across implementations.

2) DEPTH-OPTIMIZED PD

Recent PD research by Larasati et al. [15] highlights the critical role of point doubling, particularly in scenarios where the scalar k results in point coincidence. three distinct PD circuit models are introduced, emphasizing the relevance of PD in scalar multiplication (kP). Figure 5(b) illustrates the PD circuit model developed as the second model design by Larasati et al., which was later identified as depth-optimized PD and elaborated by Wardhani et al. [7] within a Qiskit-based environment and in this work implemented as a combination of PA and PD operations —particularly in the context of binary elliptic curve cryptanalysis. The quantum cost comparison results are illustrated in the Point Doubling curve analysis, as shown in Figure 2(e).

C. LEVEL-3 POINT MULTIPLICATION

While several point multiplication design references for point multiplication exist in classical hardware computing—as discussed in [18], [19], and [20],—detailed explorations in quantum environments remain limited. Therefore, in a quantum environment, this study adopts the point multiplication approach proposed by prior work (e.g., [5], [7]), which further analyzed a depth-efficient ECPM circuit.

In this work, we review and summarize various improvements in the field of quantum cryptanalysis by organizing them into a hierarchical structure elliptic curve, as illustrated in Figure 1, including for point multiplication stage. This structure serves as the foundation for iterative quantum cryptanalysis circuit designs based on Shor’s algorithm, following the approach introduced by Roetteler et al., for solving the Elliptic Curve Discrete Logarithm Problem (ECDLP). Specifically, let $P \in E(\mathbb{F}_p)$ be a fixed generator of a cyclic subgroup of $E(\mathbb{F}_p)$ with known order $\text{ord}(P) = r$, and let $Q \in \langle P \rangle$ be a fixed element of the subgroup generated by P . The objective is to determine the unique integer $m \in \{1, \dots, r\}$ such that $Q = [m]P$.

Shor’s algorithm proceeds through several steps. The initialization is performed by preparing two quantum registers, each of size $n + 1$ qubits, all initialized to the state $|0\rangle$. Next, the Hadamard transformation is applied to each qubit, placing the system in a uniform superposition state:

$$\frac{1}{2^{n+1}} \sum_{k, \ell=0}^{2^{n+1}-1} |k, \ell\rangle.$$

Then, in the function evaluation step (controlled elliptic curve addition), for each pair (k, ℓ) , the quantum circuit computes $[k]P + [\ell]Q$ and stores the result in a third register, yielding:

$$\frac{1}{2^{n+1}} \sum_{k, \ell=0}^{2^{n+1}-1} |k, \ell\rangle |[k]P + [\ell]Q\rangle.$$

After the function is evaluated, the third register is discarded, as it is no longer needed for the subsequent steps. Then, the Quantum Fourier Transform (QFT) is applied to the first two registers. Finally, quantum measurement is performed on these registers, followed by classical post-processing to extract the discrete logarithm m based on the measurement outcomes.

In Shor’s quantum algorithm for elliptic curves, the computation of $[k]P + [\ell]Q$ must preserve coherence across a quantum superposition. According to [10], a standard approach classically prepares powers of two (e.g., $2^i P$) and performs controlled point additions based on the binary decomposition of the scalar k . For an n -bit k , this yields a coherent superposition of all possible scalar multiplications $[k]P$ using $n + 1$ point additions.

However, recent works (e.g., [6], [7]) have investigated the viability of constructing quantum circuits based on combinations of PD and PA modules to optimize resource efficiency. Our work extends this approach by proposing a method that uses n PDs and 1 PA for computing $[k]P$, aiming to reduce depth and qubit usage while preserving the essential coherence required by Shor’s algorithm. This modified structure is validated through simulation and analysis, which still enables the generation of the full superposition state.

These works provide a systematic comparison of resource requirements for optimal PA and PD models, evaluating key metrics such as qubit count, circuit depth and width, total gate count, Toffoli (CCX) gates, and CNOT (CX) gates. Figure 6 illustrates the integration of PD and PA operations within a Shor-based quantum cryptanalysis circuit, with the overall scheme adapted in Figure 6 from [7], which builds upon prior implementations from [5], [10], and [12].

In particular, the work leverages recent advancements in quantum cryptanalysis of binary elliptic curves—especially in finite field arithmetic over $\text{GF}(2^m)$ —such as optimized Karatsuba multiplication algorithms [5], FLT-based inversion techniques [4], and alternative PA-only methods [5]. Wardhani et al. [7] present a comprehensive table comparing the resource requirements across various ECPM implementation strategies, including single-step and total-step ECPM.

The total-step ECPM method, comprising $2n$ PD + 2 PA operations, is contrasted with the PA-only method, which performs $2n + 2$ PA operations on n -bit elliptic curves. Wardhani et al. [7] also provide a detailed breakdown at the CNOT gate level and present interpolation estimates for various quantum cost considerations. In this paper, we focus specifically on minimizing qubit count, circuit depth, and overall circuit width.

A key distinction in our ECPM implementation is the consistent use of $2n$ PD + 2 PA operations, wherein the PA modules are implemented using an out-of-place variant (PA_{outplace}). The schematic representation of the ECPM circuit employed in this study is shown in Figure 2(f).

IV. RESULTS

A. QUANTUM COST AND COMPARISON

This section presents the quantum cost results and comparative analysis within a hierarchical framework for quantum cryptanalysis implementations targeting binary elliptic curve circuit designs. It highlights both established methodologies and optimization strategies in field arithmetic over $\text{GF}(2^m)$, while also introducing proposed enhancements at each hierarchical level. The discussion is contextualized within recent advancements in quantum circuit design, particularly as applied to Shor's algorithm for solving the ECDLP over binary fields. Our focus is on a systematic comparison of the key contributions by Banegas et al. [12], Putranto et al. [5], Taguchi et al. [13], and Jang et al. [14]. To enable a coherent and meaningful comparison—particularly in terms of interpolation strategies and quantum cost efficiency—results from these studies are synthesized into a unified visual representation, as illustrated in Figure 7.

In level 1 implementation, Figure 7 details the performance metrics of the improved FLT-based inversion (Algorithm 1), enhanced with Putranto et al.'s optimized Level 1 Karatsuba multiplication scheme [5]. This visualization highlights critical quantum resource indicators such as Toffoli gate count, circuit depth, qubit usage, and CNOT operations. Alongside our Level 1 optimization, we incorporate

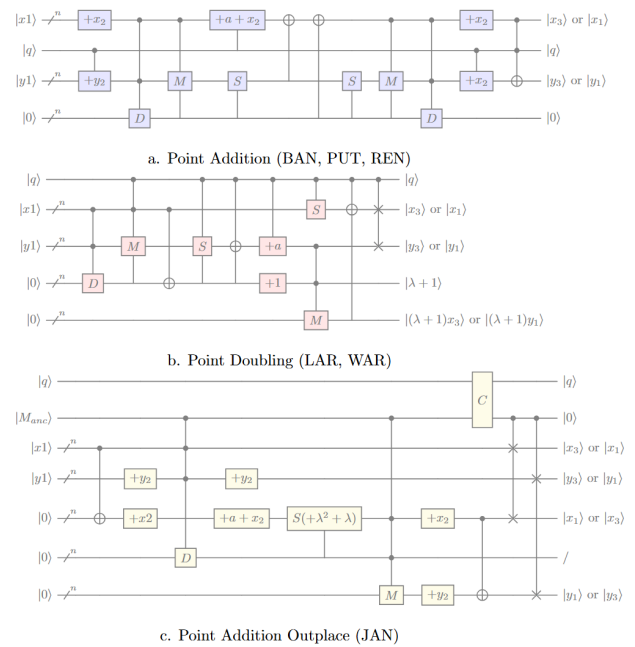


FIGURE 5. Elaborated Level-2 Quantum Circuit. (a) Point Addition (PA) circuits proposed by Banegas et al. (BAN) [12], Putranto et al. (PUT) [5], [6], and Taguchi et al. (REN) [13]. (b) Point Doubling (PD) Model 2 circuits from Larasati et al. (LAR) [15] and Wardhani et al. (WAR) [7]. (c) Point Addition circuit using an out-of-place design by Jang et al. (JAN) [14].

benchmarking FLT- and GCD-based inversion results from Banegas et al. [12]. The variants are reported with their respective depth, qubit, CNOT, and Toffoli counts.

To further enrich the comparative arithmetic level-1 landscape, the figure includes results from Taguchi et al. [13], who proposed two distinct FLT-based designs—basic and extended. Their quantum cost metrics, including qubit requirements, CNOT, and Toffoli counts, are presented accordingly. Moreover, we feature the latest contribution by Jang et al. [14], which introduces novel Level 1 designs encompassing both in-place and out-of-place circuit implementations. All these results are visualized collectively in Figure 7, enabling a comprehensive comparison of Level 1 design efficiency across multiple approaches.

At Level 2, Banegas et al. [12] laid the groundwork with their foundational PA circuits, which have since been refined by subsequent studies. Putranto et al. [5] introduced an advanced optimization technique that significantly reduces circuit depth while maintaining comparable qubit and gate costs. Wardhani et al. [7] proposed a depth-optimized design leveraging not only PA but also PD utilization computation to reduce circuit width. Meanwhile, Jang et al. [14] presented the latest contributions, offering a depth-optimized solution featuring both in-place and out-of-place variants.

The analysis shifts focus to quantum circuit implementations of elliptic curve group operations, specifically PA and PD. These operations form the core of elliptic curve arithmetic and play a crucial role in the efficiency of scalar multiplication and, by extension, quantum cryptanalysis.

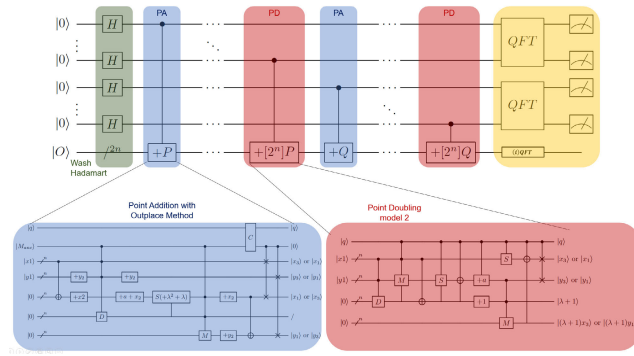


FIGURE 6. PA and PD Utilization in Shor-based Quantum Cryptanalysis Circuits. The quantum circuit adapted from Wardhani et al. [7],—originates from the foundational quantum cryptanalysis circuits design proposed by Roetteler et al. [10].

Figure 7 captures comparative insights from four notable circuit design results: Banegas et al. [12], Putranto et al. [5], Wardhani et al. [7], and Jang et al. [14]. Each work contributes to the optimization of PA and PD circuits, targeting either depth reduction, qubit minimization, or overall gate count efficiency. The comparative visualization in Figure 7 highlights the performance trade-offs among these approaches. This cross-sectional analysis underscores the evolution of Level 2 circuit design and its impact on the practical scalability of quantum cryptanalysis against binary elliptic curves.

At Level 3, which concerns point multiplication group operations—specifically PA and PD—we utilized four notable comparative designs from the works of Banegas et al. [12], Putranto et al. [5], Wardhani et al. [7], and Jang et al. [14]. Our focus is particularly on the use of $(2n + 2)$ PA and $2n$ PD + 2 PA operations within a full iteration of a Shor-based quantum cryptanalysis circuit targeting binary elliptic curves. The PA structure utilized in this work—aimed at enhancing quantum cryptanalysis of binary elliptic curves—is based on the design out-of-place proposed by Jang et al. [14].

Table 3 provides a comparative overview of quantum resource requirements as evaluated in this study. The degree-size parameters n are provided in column (i), covering a range of common security levels such as $n=8, 16, 127, 163, 233, 283$, and 571. The quantum resource metrics under review include qubit count, Toffoli gate usage, CNOT gate operations, and overall circuit depth—offering insight into both spatial (qubit-based) and temporal (depth-based) efficiency. This thorough comparison aims to benefit in identifying the most effective configurations regarding resource trade-offs for the practical implementation of space- and time-efficient ECPM circuits.

The upper portion of the table highlights Level-3 ECPM operations by integrating design strategies from prior works (Figure 5), with each approach color-coded for clarity: Banegas et al. (BAN) [12] in green, Putranto et al. (PUT) [5] in purple, Wardhani et al. (WAR) [7] in red,

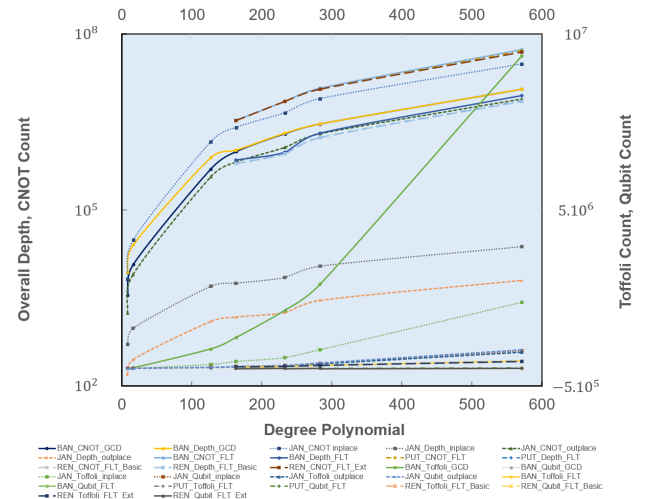


FIGURE 7. Depth Comparative Analysis. The figure presents an interpolated analysis of the parameter n and the corresponding total depth required for resource estimation, facilitating a comprehensive comparison of Level-1 circuit designs. The comparison includes FLT-based inversion with Karatsuba-depth optimization [5], GCD-based and FLT-based approaches [12], basic and extended FLT-based inversion techniques [13], as well as in-place versus out-of-place arithmetic strategies [14].

and Jang et al. (JAN) [14] in yellow. These configurations include single-step ECPM implementations, considering either a PA-only strategy or a hybrid approach that integrates both PA and PD components. The bottom portion of Table 3 extends the evaluation to total resource usage when each referenced approach is applied within a single iteration of Shor's algorithm for ECDLP, following the circuit structure outlined by Roetteler et al. [10].

B. DECOMPOSING MULTI-CONTROLLED QUANTUM GATES

In practical quantum implementations, hardware constraints frequently hinder the direct application of gates operating on multiple qubits, particularly those involving several control qubits, like multi-controlled Toffoli gates. To adhere to these constraints, such gates must be decomposed into sequences of one- and two-qubit gates, potentially resulting in considerable overhead in both gate depth and qubit utilization [36]. Consequently, numerous circuit designers either eschew these gate types altogether or devise more efficient methods for their decomposition and optimization.

In accordance with these initiatives, Ufrecht et al. [37] propose a technique for diminishing the complexity of multi-controlled gates by partitioning circuits into smaller, manageable substructures. Their methodology, aptly designed for the constraints of noisy intermediate-scale quantum (NISQ) devices, diverges from previous circuit-cutting strategies that emphasized the segmentation of qubit wires or two-qubit interactions. Their method specifically focuses on slicing multi-controlled operations and is based on the principles of ZX-calculus, a graphical formalism for quantum mechanics.

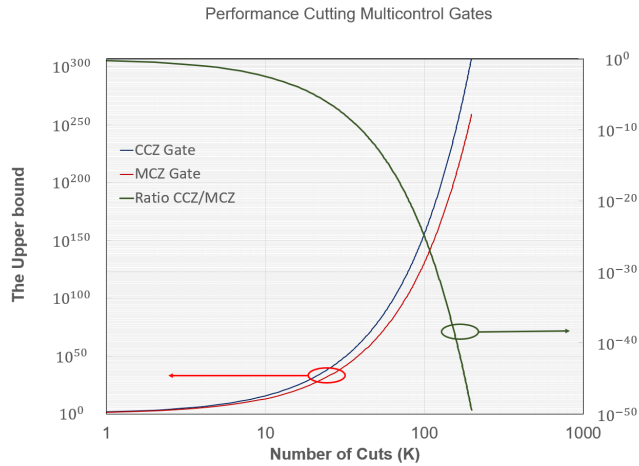


FIGURE 8. Decomposing multi-controlled Quantum Gates We demonstrate slicing multi-controlled gates method proposed by Ufrecht et al. significantly reduces the circuit size, lowering it from an upper bound of $O(6^{2K})$ (visualized in blue) to a more compact structure with a complexity of $O(4.5^{2K})$ (shown in red), where K denotes the number of applied cuts. The resulting efficiency improvements are highlighted with a green line that represents the ratio of gate cuts to size reduction.

Figure 8 illustrates the enhancement in circuit efficiency achieved by Ufrecht et al.'s slicing method: the initial circuit complexity, capped at $O(6^{2K})$ (blue), is diminished to $O(4.5^{2K})$ (red) via slicing, with K representing the number of slicing points. The efficiency enhancement is illustrated by the green trendline, signifying the reduction ratio attained via gate slicing.

This ZX-calculus-based strategy is particularly relevant to our study, as several circuits we examine—including the PD architecture by Larasati et al. [15] and the advanced out-of-place PA circuit by Jang et al. [14]—make extensive use of multi-controlled gates. This is evident in the quantum circuit examples shown in Figure 5, parts (b) and (c), which contrast with the PA circuit in part (a) that avoids multi-controlled gates, aligning more closely with standard quantum circuit designs. The presence of multi-controlled gates in parts (b) and (c) implies potentially more complex decompositions into elementary gates and warrants further elaboration on the associated arithmetic operations.

In our resource analysis, we include the “Multi” column in Table 3 to estimate the quantum cost overhead introduced by multi-controlled gate usage. Specifically, we assume that the M_{anc} register used in Jang et al.'s circuit functions as a control component. As described in Algorithm 1, Step 3 of Jang et al. [14], the M_{anc} contents are control bits derived from q , which govern the CTRL_CONST_ADD operation and produce the output y . This output y , in turn, influences the λ parameter used in the D module. Consequently, the coupling with ancilla qubits—as embodied by M_{anc} —is explicitly considered in our resource estimation, as it can significantly affect the circuit's depth and complexity in the targeted quantum architecture.

C. SPACE-AND TIME-EFFICIENT APPROACH

The focus of contemporary quantum cryptanalysis research has shifted toward resource efficiency. The quantum circuit depth denotes the number of sequential layers of gates that must be executed, considering gates that can function concurrently. Given that quantum gate operations necessitate time, circuit depth is a crucial metric for assessing the total execution time. In addition to circuit width, depth is a crucial metric for assessing the feasibility and resource demands of executing a quantum circuit, alongside an indicator of spatial complexity, particularly qubit count. In this study, we interpret circuit width as a measure of spatial complexity (i.e., qubit count), whereas depth serves as an estimate of temporal complexity. While these benchmarks are broadly recognized, the efficiency of quantum circuits is further affected by factors such as gate fidelity, connectivity limitations, and error correction overhead, requiring further investigation.

Wardhani et al. [7] introduced the terms single step and total step in their analysis of quantum circuits for binary elliptic curve cryptanalysis to distinguish between different layers of resource evaluation. A single step refers to an individual ECPM circuit composed of PA or PA/PD operations, incorporating advanced Karatsuba multiplication and optimized FLT-based inversion techniques. In contrast, the total step pertains to a full ECPM (Elliptic Curve Point Multiplication) implementation—comprising $2n$ PD + 2 PA operations—representing the overall resource demand for a complete point multiplication cycle under the Shor-based quantum circuit, in our case quantum cryptanalysis for binary elliptic curve. In this work, we adopt the same definitions: the single step quantifies the quantum resource cost of a single PA or PD operation, while the total step represents the cumulative resource estimation required to execute a full iteration of Shor-based quantum cryptanalysis on binary elliptic curves. This analysis is conducted using three ECPM construction strategies: approach $(2n + 2)$ PA [5], [12], approach $2n$ PD + 2 PA [7], and our proposed method, approach $2n$ PD + 2 PA_{outplace}.

1) SINGLE-STEP

This evaluation presents a thorough and organized comparison of quantum resource trade-offs among various implementations, as illustrated in Table 3, which contrasts multiple designs at both the single-step and total-step levels. The upper section of columns (ii)–(v) in Table 3 displays the single-step qubit count result assessments obtained from each circuit design—BAN [12], PUT [5], WAR [7], and JAN [14]—to enable a uniform comparison among the different models. The table presents a comparative analysis of qubit consumption across four elliptic curve quantum circuit implementations—BAN, PUT, WAR, and JAN—evaluated over varying degrees $n = 8, 16, 127, 163, 233, 283, 409$, and 571. As a result, all implementations exhibit a growth

TABLE 3. Quantum Cost and Resource Comparison. The table presents the outcomes of the Qiskit environment simulation and resource estimation in terms of the detailed analysis of Toffoli and CNOT gates, qubits, and circuit depth, along with the point addition results developed by Banegas et al. (BAN) [12] in green, Putranto et al. (PUT) [5] in purple, Wardhani et al. (WAR) [7] in red, Jang et al. (JAN) [14] in yellow. Columns ii–v of the qubit comparison table present estimates derived from the Qiskit simulations conducted in this study. The subsequent column in the upper section of the table illustrates the single-step values reported in prior works by BAN, PUT, WAR, and JAN, while the lower section presents the corresponding total-step values as well as those from our proposed approach.

n	qubits				BAN (Single-step PA)			PUT(Single-step PA)			WAR (Single-step ECPM)			JAN (Single-step PA out)		
	BAN	PUT	WAR	JAN	Toffoli	CNOT	Depth	Toffoli	CNOT	Depth	Toffoli	CNOT	Depth	Toffoli	CNOT	Depth
i	ii	iii	iv	v	vi			vii			viii			ix		
8	68	74	89	241	7,360	3,522	8,562	348	2,734	210	205	1,764	23	178	1,761	159
16	125	194	225	859	21,016	11,686	25,205	1,344	15,924	623	761	9,688	39	680	7,953	283
127	904	2,320	2,795	33,157	559,141	497,957	776,234	57,143	1,352,497	7,010	85,469	2,379,554	261	28,659	370,120	1,267
163	1,157	2,805	3,424	57,521	893,585	827,623	1,262,280	96,299	2,137,063	14,632	52,970	1,260,909	333	48,583	650,277	1,495
233	1,647	4,228	5,127	89,222	1,669,299	1,615,287	2,406,230	152,067	4,480,745	46,702	82,665	2,621,087	473	76,342	1,158,949	1,800
283	1,998	5,694	6,793	154,945	2,427,369	2,359,187	3,503,964	267,115	7,376,571	34,792	144,388	4,244,506	573	134,115	2,007,399	2,902
571	4,015	13,167	15,418	531,621	8,987,401	9,081,061	13,238,554	935,883	32,888,178	93,142	499,878	18,614,355	1,149	468,707	7,833,731	6,313
n	BAN ((2n + 2) PA)				PUT ((2n + 2) PA)			WAR (2nPD + 2PA)			JAN ((2n + 2) PA)			Our (2n PD + 2 PA _{OUTPLACE})		
	Toffoli	CNOT			Toffoli	CNOT		Toffoli	CNOT	Multi	Toffoli	CNOT	Multi	Toffoli	CNOT	Multi
i	x				xi			xii			xiii			xiv		
8	132,480		858,276		6,264	86,796		3,976	55,608	36	3,204	31,698	72	3,690	31,752	40
16	714,544		4,684,588		491,632	815,592		27,040	14,776,984	68	23,120	270,402	136	25,874	329,392	72
127	143,140,096		986,317,568		13,509,888	427,810,560		21,823,412	74,010,296	512	7,336,704	94,750,720	1,024	21,880,064	609,165,824	516
163	293,095,880		2,030,035,624		31,586,072	890,473,096		17,460,946	518,343,828	656	15,935,224	213,290,856	1,312	17,374,160	413,578,152	660
233	781,231,932		5,443,345,908		71,167,356	2,523,992,796		38,826,088	1,459,625,244	936	35,728,056	542,388,132	1,872	38,687,220	1,226,668,716	940
283	1,378,745,592		9,612,491,768		151,721,320	5,100,220,248		82,257,966	2,904,427,204	1,136	76,177,320	1,140,202,632	2,272	82,012,384	2,410,879,408	1,140
571	10,281,586,744		72,078,254,248		1,070,650,152	44,047,976,544		572,732,570	24,731,217,540	2,288	536,200,808	8,961,788,264	4,576	571,860,432	21,294,822,120	2,292

in qubit requirements as n increases, but the rate and scale of increase vary significantly across the designs. The qubit counts across BAN, PUT, and WAR remain relatively low and within a comparable range, while JAN demonstrates a noticeably higher footprint.

Table 3 provides a detailed comparison of single-step quantum arithmetic implementations for various circuit designs—BAN, PUT, WAR, and JAN—evaluated over increasing input sizes. Our single-step results implementation analysis is presented in columns (vi)–(ix), which enumerate the counts of Toffoli gates, CNOT gates, and circuit depth for each method, with particular emphasis on Level-3 ECPM operations. Analysis of qubit columns reveals that the design by Banegas et al. [12] provides the most qubit-efficient (space-efficient) implementation; nonetheless, it demonstrates the greatest circuit depth. Conversely, the design by Wardhani et al. [7] attains the minimal depth (i.e., maximal time efficiency), albeit necessitating a greater number of qubits than the Banegas et al. method, thereby situating its space cost between the designs put forth by Putranto et al. [5] and Jang et al. [14].

2) TOTAL-STEP WITH ECPM ((2N + 2) PA AND 2N PD + 2 PA)

This study expands resource estimations to full-step implementations, specifically a full iteration of Shor-based quantum cryptanalysis for binary elliptic curves, adhering to the circuit architecture proposed by Roetteler et al. [10], thereby providing a more comprehensive analysis beyond the Level-3 evaluation. Column (x) delineates the methodology employed by Banegas et al., which posits that a complete iteration necessitates $(2n + 2)$ PA operations. Column (xi) similarly illustrates the depth-optimized methodology employed by Putranto et al., which utilizes the identical estimate of $(2n + 2)$ PA. Column (xii) presents the findings of Wardhani et al., who advocate for a more modular methodology employing $2n$ PD + 2 PA operations per iteration. Column (xiii) displays

the detailed findings of Jang et al. [14], who likewise posit a total of $(2n + 2)$ PA operations, in agreement with Banegas et al. and Putranto et al.

3) TOTAL-STEP WITH ECPM (2N PD + 2 PA_{OUTPLACE})

Table 3 summarizes the total-step quantum resource requirements of four ECPM schemes—BAN, PUT, WAR, and JAN—evaluated across polynomial degrees ranging from 8 to 571. The reported metrics include Toffoli and CNOT gate counts, along with the number of multiplications performed. Among the four full-step implementations under the total-step ECPM strategy—namely, the $(2n + 2)$ PA and $2n$ PD + 2 PA configurations—Jang et al.’s design, as detailed in [14], achieves the lowest Toffoli and CNOT gate counts. However, this optimization heavily relies on the use of multi-controlled gates, whose practical efficiency may degrade in large-scale quantum circuits or hardware-oriented implementations, as discussed in [37]. To reflect this, Table 3 includes a “Multi” column, which specifies the number of multi-controlled gates—a critical factor contributing to overall circuit complexity.

As shown in Table 3, Jang et al.’s design exhibits a significantly higher number of multi-controlled gates compared to the other works, particularly when contrasted with the design by Wardhani et al. [7]. While Wardhani et al.’s implementation does not achieve the lowest gate count, their design—evaluated in both single-step and total-step configurations—demonstrates minimal circuit depth (i.e., maximal time efficiency), though it requires more qubits than the approach by Banegas et al. [12]. Nonetheless, Wardhani et al.’s approach offers a spatial cost close to that of Jang et al. [14], thus striking a meaningful balance in quantum resource utilization.

This study emphasizes that achieving a design that is both space- and time-efficient requires the careful evaluation of both metrics. In quantum computing, circuit width (qubit count) and depth are fundamental indicators of resource efficiency. While qubit count reflects the spatial

resource demands, circuit depth denotes the number of sequential steps required to execute the computation. Since each gate operation incurs a finite delay, depth directly influences total runtime, serving as a crucial measure of time complexity and scalability. Although quantum hardware is evolving to accommodate larger qubit counts, minimizing circuit depth and gate count remains essential for practical and fault-tolerant implementations.

To address this trade-off, we report quantum cost results for a modified $2n$ PD + 2 PA_{outplace} configuration that integrates the out-of-place PA module from Jang et al. with the PD strategy employed by Wardhani et al. [7]. The outcome of this integration is presented in the white column (xiv) of the comparison table. This configuration achieves a significantly reduced count of multi-controlled gates—comparable to Wardhani et al.—while also yielding competitive Toffoli gate counts and circuit depth.

V. CONCLUSION

This work presented a structured and resource-aware framework for implementing quantum cryptanalysis of binary elliptic curves over the finite field $\text{GF}(2^m)$. Drawing upon key contributions from prior studies, including [5], [7], [12], [13], [15], and [14], the analysis integrated simulation-based evaluations and numerical cost estimation to offer actionable insights into the design of quantum cryptanalytic circuits that achieve a balanced trade-off between spatial (qubit count) and temporal (circuit depth) efficiency. In particular, the investigation explored multiple circuit construction strategies tailored to optimize Shor-based quantum cryptanalysis binary elliptic curve cryptography.

This study proposed improvement strategies for quantum circuit design within the framework of Shor's algorithm for solving the elliptic curve discrete logarithm problem (ECDLP) over binary fields—specifically by leveraging the ECPM structure from Wardhani et al. and the out-of-place PA method from Jang et al. While the design by Jang et al. demonstrates strong efficiency through minimal Toffoli and CNOT usage, it comes with increased complexity due to the use of multi-controlled gates, potentially escalating practical computational costs. To address this, the proposed design—based on a $2n$ PD + 2 PA_{outplace} configuration—incorporates advanced multi-controlled gate decomposition techniques, building upon the ECPM structure $2n$ PD + 2 PA from Wardhani et al. and the out-of-place PA design by Jang et al. This enhancement preserves the saving advantages of Jang et al.'s approach while significantly reducing multi-control gate overhead, thereby contributing to more practical and scalable implementations of quantum cryptanalysis.

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